Testing Kepler's Third Law with Known Exoplanets

How closely do reported exoplanet orbital parameters agree with Kepler's third law?

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Introduction

Kepler's third law relates a planet's orbital period to the size of its orbit. The same law applies to planets outside our Solar System, but their host stars do not all have the same mass. This means that orbital distance alone is not enough to describe the period. When the planet's mass is much smaller than that of its star, the relation can be written as (Tatum, n.d.)

$$P^2 \simeq \frac{a^3}{M_\star}, \tag{1}$$

where P is measured in years, a is the semi-major axis in astronomical units (AU), and $M_\star$ is the stellar mass in solar masses. In this project I compared the orbital parameters of known exoplanets with Kepler's prediction. I first looked at the relation between period and distance, and then included the mass of the host star.

Data sample

The data came from the NASA Exoplanet Archive's *Planetary Systems Composite Parameters* table (PSCompPars). I downloaded the CSV through the archive's TAP service, selecting the planet name (`pl_name`), orbital period (`pl_orbper`), semi-major axis (`pl_orbsmax`), and host-star mass (`st_mass`) (NASA Exoplanet Archive, 2026). These are reported in days, AU, and solar masses, respectively. The downloaded file contained 6,375 entries.

I used pandas to remove entries with missing values in any of the required columns. This removed 774 entries. The code also checked for non-positive values and duplicate planet names, but neither check removed any additional rows. In total, **5,601 planets** remained for the analysis.

Data analysis

The orbital periods were converted from days to years using 365.25 days per year. I then took base-10 logarithms to turn the power-law relation into a straight-line fit. Two regression models were used:

$$\log_{10} P = b_0 + b_a \log_{10} a, \tag{2}$$
$$\log_{10} P = c_0 + c_a \log_{10} a + c_M \log_{10} M_\star. \tag{3}$$

The first model considers only orbital distance, with an expected slope of 1.5. The second includes stellar mass, giving expected coefficients of +1.5 for $\log_{10} a$ and −0.5 for $\log_{10} M_\star$. I fitted the models using NumPy and compared them with R^2 and root mean squared error (RMSE). Both measures were calculated from $\log_{10} P$, so the RMSE is not in years. I also plotted the residuals (observed minus fitted $\log_{10} P$) for the simple model.

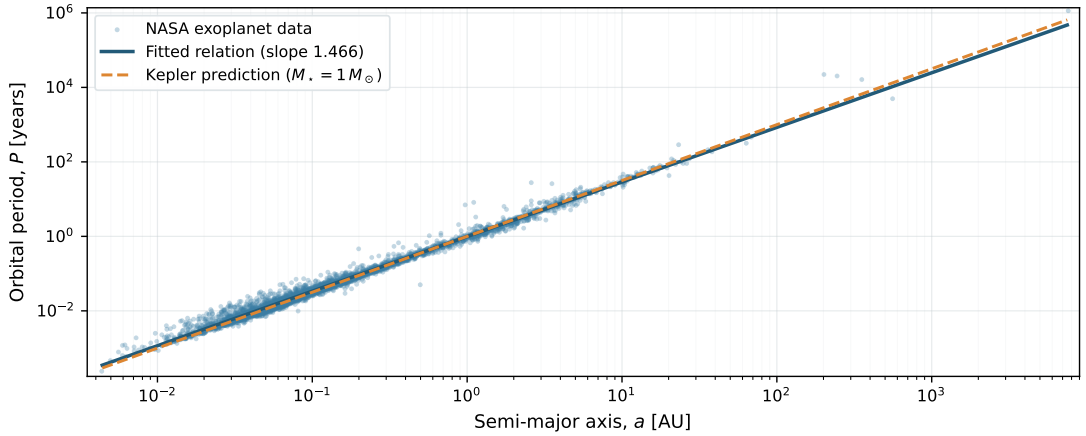


Figure 1: Orbital period against semi-major axis for the 5,601 planets on logarithmic axes. The solid line is the simple regression; the dashed line shows Kepler's prediction for a one-solar-mass star.

Results and discussion

As Figure 1 shows, orbital period increases strongly with semi-major axis. The simple regression gave a slope of **1.466**, only 0.034 below the predicted value of 1.500. The fit had an R^2 of 0.990, although there was still scatter around the line.

When host-star mass was included, the fitted coefficients became **1.497** for $\log_{10} a$ and −0.474 for $\log_{10} M_\star$. Both are close to the expected values. The RMSE decreased from 0.093 to 0.035, a reduction of about **62%**. Table 1 gives an overview of the two models.

Table 1: Results of the two regressions using the same 5,601 planets. RMSE is in $\log_{10} P$ units.

Quantity	Simple fit	Mass-aware fit	Expected
Coefficient of $\log_{10} a$	1.466	1.497	1.500
Coefficient of $\log_{10} M_\star$	—	−0.474	−0.500
R^2	0.990	0.999	—
RMSE (log-period)	0.093	0.035	—

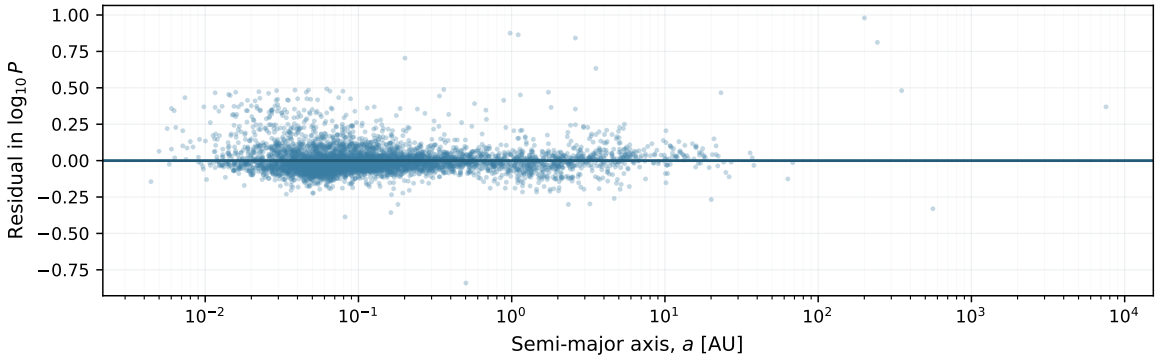


Figure 2: Residuals of the *simple* regression, plotted against semi-major axis. Most lie near zero, with a few larger outliers.

The improvement makes sense physically. At the same orbital distance, a planet around a more massive star has a shorter period. Leaving stellar mass out therefore adds variation that the simple model cannot account for. Figure 2 shows the residuals of that *simple* fit, not the mass-aware one. The lower RMSE tells us that including mass brings the fitted periods closer to the reported periods overall.

There is an important limitation, though. The PSCompPars table combines parameters from published studies, sometimes from different sources (NASA Exoplanet Archive, 2023). Some semi-major axes can also be derived from the orbital period and stellar mass using Kepler's law itself; NASA describes this explicitly for its Kepler Objects of Interest data (NASA Exoplanet Archive, n.d.). Because of this, the close fit mainly shows that the reported catalogue values are *consistent* with Kepler's law. It should not be taken as a fully independent experimental test.

Conclusion

The reported orbital parameters of the 5,601 exoplanets agree very closely with Kepler's third law. The simple model gave an orbital-distance exponent of 1.466, compared with the predicted 1.500. When stellar mass was included, the fitted exponents were 1.497 for distance and −0.474 for stellar mass, both close to theory. The RMSE also decreased from 0.093 to 0.035. Overall, this answers the research question: Kepler's law describes the relationships in the NASA catalogue very well, particularly when the different host-star masses are taken into account. Still, since some semi-major axes may already have been calculated using the law, this is mainly a check of how consistent the catalogue values are, rather than an independent test.

References

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